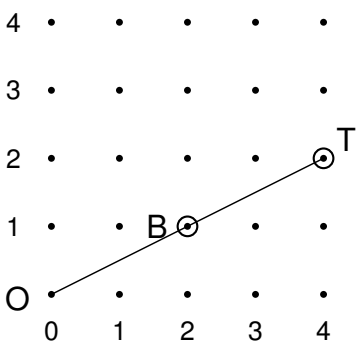
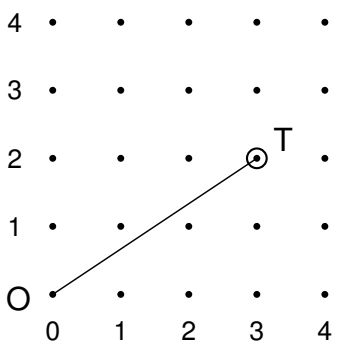
Stretch a thread from O to another dot T. A dot exactly on the thread between O and T blocks T. B marks a blocker. Keep O fixed while you test.

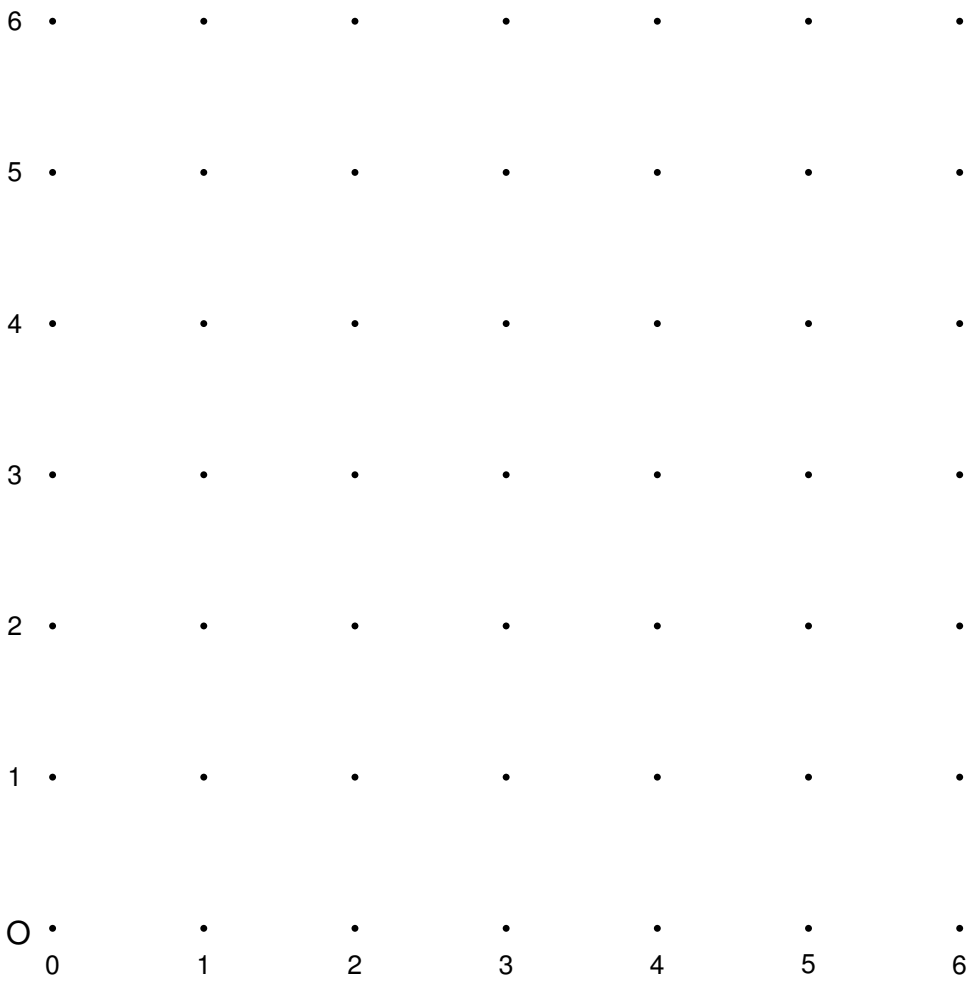


hidden

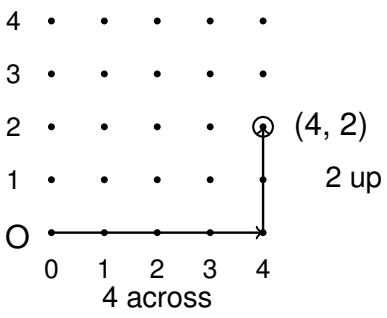


visible

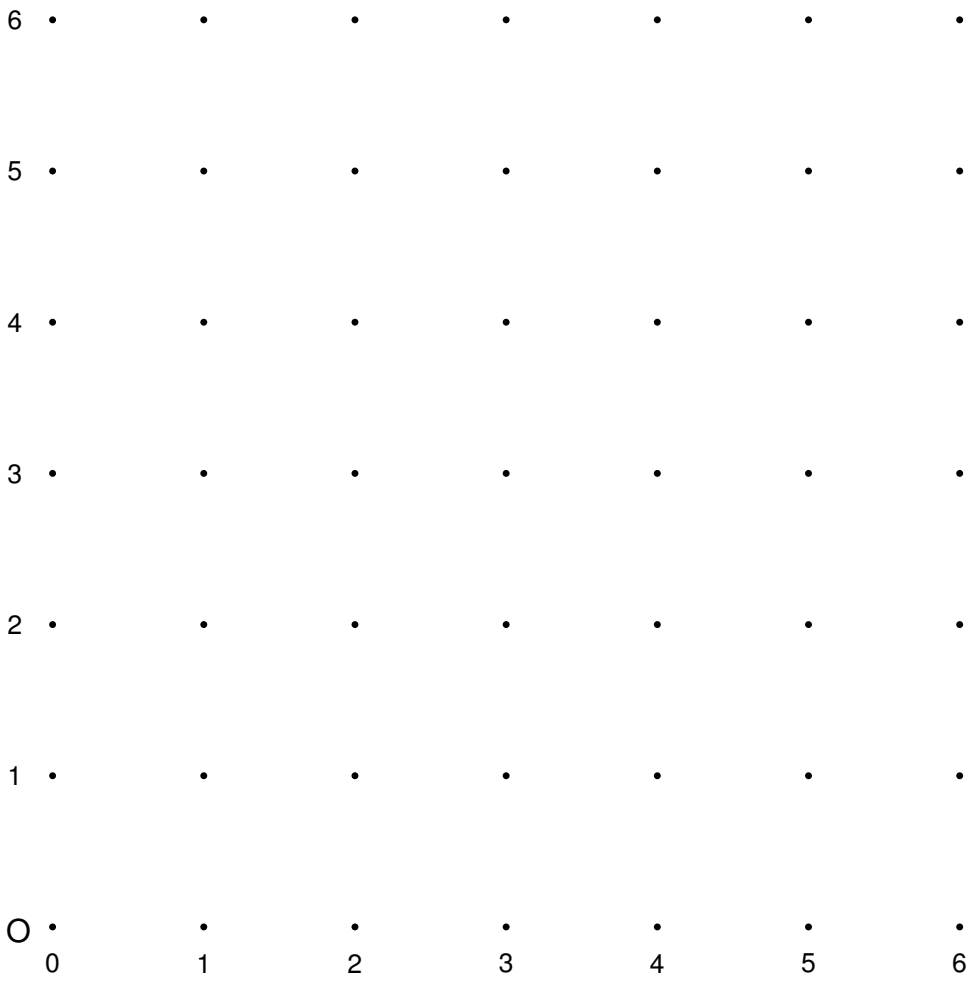
Problem 1: Circle every dot visible from O. For three hidden dots, mark their first blocker.



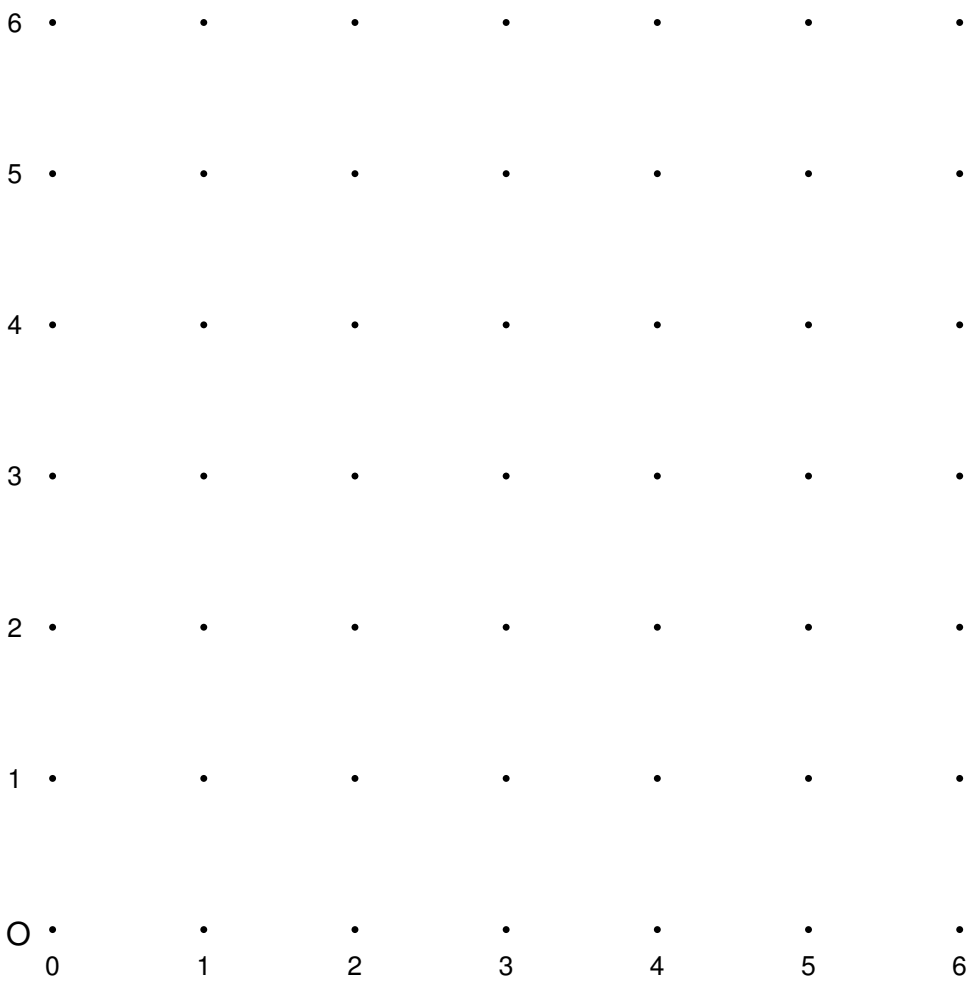
(4, 2) means 4 across and 2 up from O.



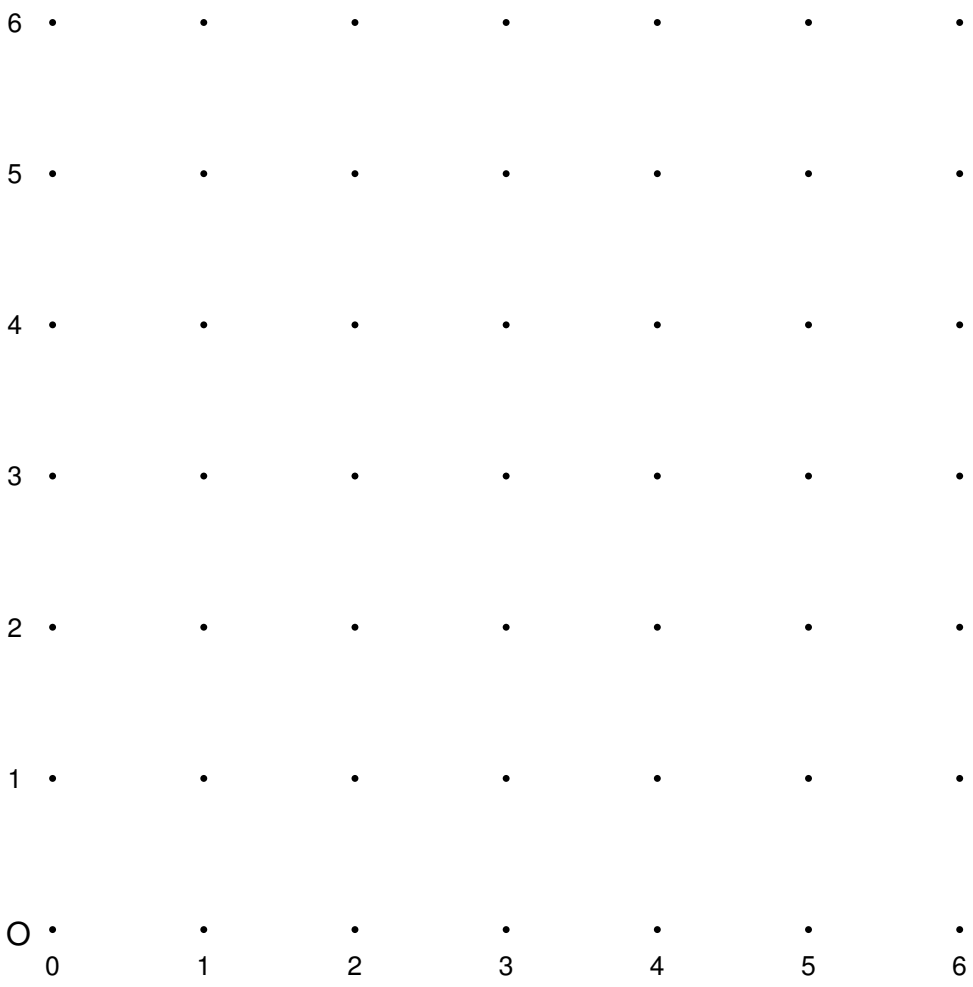
Problem 2: Test these targets: (6, 4), (6, 3), (5, 3), (5, 2), (4, 4) and (4, 1). Find every dot between O and each target.



Problem 3: Find all targets on this grid with exactly one blocker. Then find all targets with exactly two blockers.



Problem 4: Which dots lie on the same straight line from O as (1, 1)? Make groups for (1, 1), (2, 1), (3, 2), (1, 2) and (1, 0).



Problem 5: Find a number rule that predicts whether a dot is visible from O. Use it to predict (12, 8), (10, 7), (15, 5) and (11, 1). Explain how the rule matches a straight thread.

Problem 6: A grid continues as far as you like. Give a row where every dot is visible, and explain why each one is visible.